Predicting Provincial Carbon Intensity Changes in China Using Machine Learning

Qingya Yang · DS & Climate Change

PROBLEM STATEMENT

China's national CO_2 intensity fell 48.4% vs. 2005 levels — yet provincial outcomes vary enormously. This project asks: **can structural and policy variables predict short-run changes in provincial carbon intensity?**

We frame this as a supervised regression problem: given province-level economic, energy, and policy characteristics in year *t−1*, predicting annual carbon intensity change (Δln CI) in year *t*.

DATA

30 Chinese provinces, 2008–2021 (390 province-year observations after lagging)

Target: Annual $\Delta\ln(CO_2/GDP)$
Split: Train 2009–2017, **Test:** 2018–2021

Features (all lagged 1 year):
• Log GDP per capita • Urbanization rate
• Secondary industry share • Tertiary industry share
• Coal share • Policy intensity index • Log carbon intensity

METHODS

Naïve Baseline
Repeat last year's CI change (no learning). Sets the minimum performance bar.

OLS Regression
Linear model. Interpretable coefficients; cannot capture variable interactions.

Ridge Regression
OLS + L2 penalty. Shrinks all coefficients to prevent overfitting. λ selected via 5-fold CV.

Lasso Regression
OLS + L1 penalty. Zeros out weak predictors: automatic variable selection. λ via 5-fold CV.

Random Forest
500 decision trees trained on random data/feature subsets; predictions averaged. Captures nonlinearity.

XGBoost
100 rounds of gradient boosting: each tree corrects prior errors. η=0.05, max_depth=4.

EVALUATION METRICS

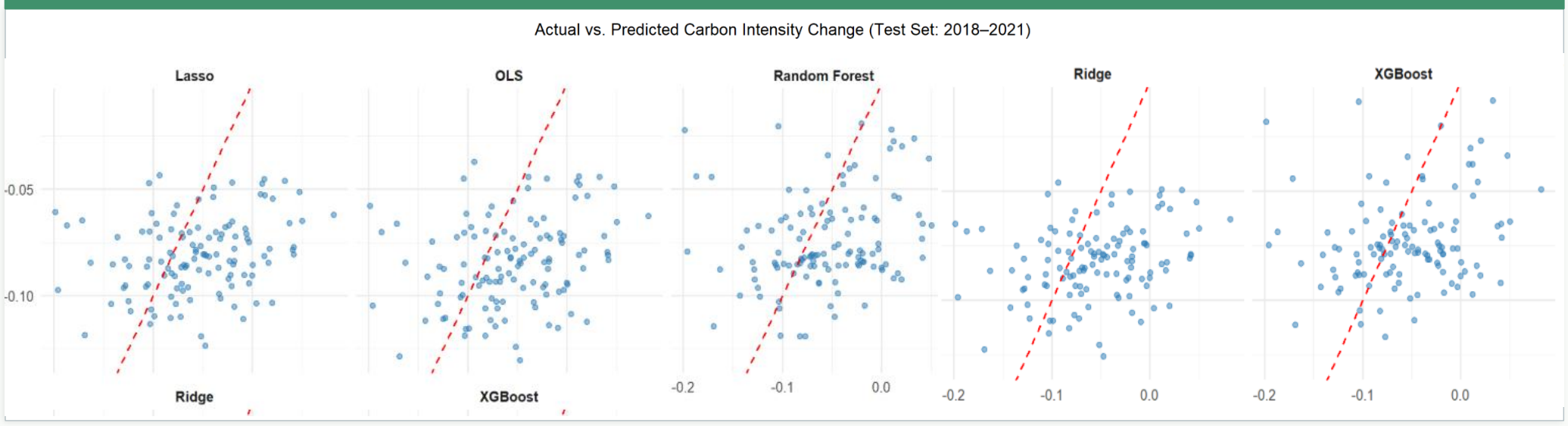
MAE (Mean Absolute Error): Average prediction error magnitude
RMSE (Root Mean Squared Error): Penalizes large errors more heavily
R^2: Share of variance in Δln CI explained by the model

MODEL PERFORMANCE (TEST SET: 2018–2021)

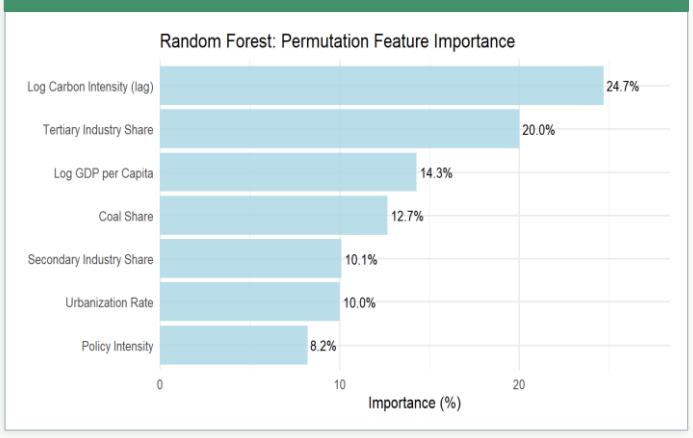
Model	MAE	RMSE	R^2
Naïve Baseline	0.0566	0.0745	0.0046
OLS	0.0500	0.0612	0.0661
Ridge	0.0487	0.0599	0.0686
Lasso	0.0484	0.0596	0.0641
RF ★	**0.0453**	**0.0579**	**0.0496**
XGBoost ★	**0.0451**	**0.0577**	**0.0420**

★ Best MAE/RMSE. All R^2 values low.

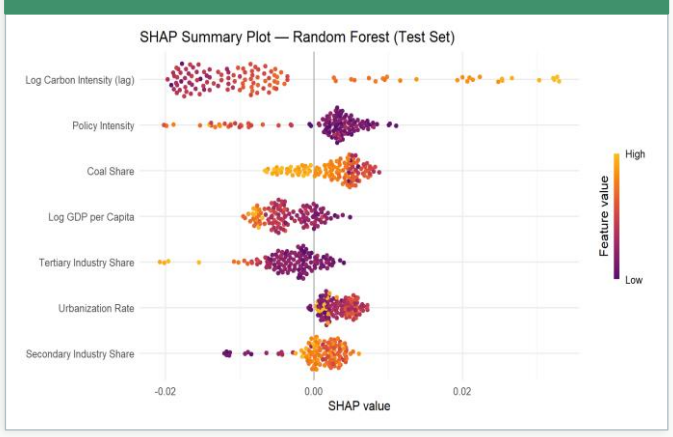
ACTUAL VS. PREDICTED (TEST SET)



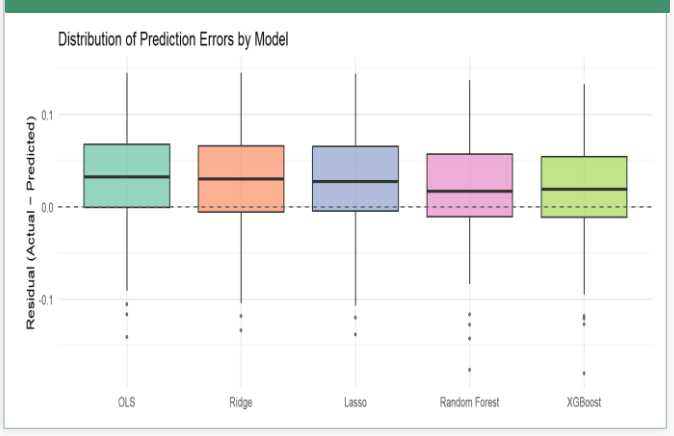
FEATURE IMPORTANCE — RANDOM FOREST



SHAP VALUES — RANDOM FOREST



PREDICTION ERROR DISTRIBUTION



DISCUSSION & KEY FINDINGS

1. Low R^2 across all models (max 0.069) reflects a fundamental limit: **structural variables explain little of year-to-year CI volatility**, which is driven by short-run shocks and policy lags.

2. Tree models (RF, XGBoost) achieve the best MAE/RMSE, confirming that nonlinear relationships matter, but R^2 is not higher than linear models.

3. Prior carbon intensity level (lag) is the strongest predictor across all models, indicating mean-reversion: high-emission provinces tend to decarbonise faster.

4. Policy Intensity is consistently the weakest predictor. Lasso dropped it; SHAP shows near-zero contributions. Structural variables dominate short-run dynamics.

LIMITATIONS

Small sample: Only 270 training observations (30 provinces × 9 years) constrain model capacity and CV stability.

Policy measurement: The policy index captures policy volume (word count/NLP), not enforcement intensity or compliance rates.

Temporal leakage risk: Even with lagged features, unobserved province-level trends may confound predictions.

Missing variables: FDI flows, technology adoption, and weather shocks are excluded but may explain a share of residual variance.

FUTURE WORK

Enforcement data: Incorporate inspection frequency, penalty records, or satellite-derived emissions to measure actual policy implementation.

Panel-aware models: Apply fixed-effects or mixed-effects extensions to explicitly account for province-level heterogeneity.

Longer horizon: Extend to 2-year and 3-year ahead predictions to capture delayed structural effects.

City-level analysis: Disaggregate to prefecture level using Dong & Wang's city-level policy dataset to increase sample size and spatial granularity.

CARBON COST OF THIS PROJECT

This project used the CodeCarbon Python library to estimate the computational footprint of model training. Training times were recorded in R via system.time() and passed to CodeCarbon for hardware-aware carbon estimation. All models were trained on a local CPU (no GPU acceleration). Total training emissions across all six models amount to just 0.0089 gCO_2e, with XGBoost being the most computationally intensive (0.23s, 0.0016 gCO_2e) and OLS the lightest (< 0.01s). These figures are negligible compared to the domain studied, demonstrating that interpretable ML pipelines for climate policy analysis can be run at near-zero carbon cost.

Data: CEADS Carbon Emissions · China Statistical Yearbooks · Dong & Wang et al. (2024) Provincial Low-Carbon Policy Intensity Dataset · Methods: R (ranger, glmnet, xgboost, shapviz, kernelshap)